Experiment No.5
Support Vector Machine (SVM) for Credit Default Prediction

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# Support Vector Machine (SVM)

# with Hyperparameter Tuning

# and Heatmap Visualizations

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# Install required libraries (run once if needed)

# pip install numpy pandas matplotlib seaborn scikit-learn

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns


from sklearn.datasets import load_breast_cancer

from sklearn.model_selection import train_test_split, GridSearchCV

from sklearn.preprocessing import StandardScaler

from sklearn.svm import SVC

from sklearn.metrics import confusion_matrix, classification_report, accuracy_score


# --------------------------------

# 1. Load Dataset (Intermediate Kaggle Level)

# --------------------------------
```

```
data = load_breast_cancer()
X = pd.DataFrame(data.data, columns=data.feature_names)
y = data.target


# --------------------------------
# 2. Train-Test Split
# --------------------------------
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)


# --------------------------------
# 3. Feature Scaling (Important for SVM)
# --------------------------------
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)


# --------------------------------
# 4. Hyperparameter Tuning
# --------------------------------
param_grid = {
    'C': [0.1, 1, 10, 100],
    'gamma': [1, 0.1, 0.01, 0.001],
    'kernel': ['rbf']
}
```

```
grid = GridSearchCV(
    SVC(),
    param_grid,
    refit=True,
    verbose=1,
    cv=5
)

grid.fit(X_train_scaled, y_train)

print("Best Parameters Found:", grid.best_params_)

# ------------------------------
# 5. Model Evaluation
# ------------------------------
best_model = grid.best_estimator_
y_pred = best_model.predict(X_test_scaled)

print("\nAccuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))

# ------------------------------
# 6. Confusion Matrix Heatmap
# ------------------------------
cm = confusion_matrix(y_test, y_pred)
```

```
plt.figure()
sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix Heatmap")
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.show()


# ------------------------------
# 7. Hyperparameter Tuning Heatmap
# ------------------------------
results = pd.DataFrame(grid.cv_results_)

pivot_table = results.pivot_table(
    values='mean_test_score',
    index='param_C',
    columns='param_gamma'
)

plt.figure()
sns.heatmap(pivot_table, annot=True, fmt=".4f")
plt.title("Hyperparameter Tuning Heatmap (C vs Gamma)")
plt.xlabel("Gamma")
plt.ylabel("C")
plt.show()
```

Here is the **modified theory section** properly written for your selected dataset:

Dataset Description

Dataset Name: Default of Credit Card Clients Dataset

Source (Kaggle): https://www.kaggle.com/datasets/uciml/default-of-credit-card-clients-dataset

Total Instances: 30,000

Total Input Features: 23

Target Variable: `default.payment.next.month`

Problem Type: Binary Classification

Class Distribution: Imbalanced (~22% default cases, ~78% non-default cases)

This dataset contains demographic information, credit data, payment history, bill statements, and previous payment records of credit card clients in Taiwan.

Mathematical Formulation of SVM

Support Vector Machine aims to find the optimal hyperplane that maximizes the margin between classes.

Linear Decision Boundary:

w^T x + b = 0

Where:

- w = weight vector
- b = bias
- x = feature vector

Optimization Objective:

$$\min \frac{1}{2}||w||^2$$

Subject to:

$$y_i(w^T x_i + b) \geq 1$$

Soft Margin (Real-world data):

$$\min \frac{1}{2}||w||^2 + C\sum \xi_i$$

Where:

- C = regularization parameter
- ξ_i = slack variables

Algorithm Limitations

- Computationally expensive for very large datasets
- Sensitive to choice of hyperparameters (C, kernel)
- Requires feature scaling
- Not ideal for extremely noisy datasets
- Hard to interpret compared to logistic regression
- Does not perform well when classes overlap heavily

Methodology / Workflow

Step-by-Step Process:

- Load dataset
- Drop unnecessary columns (ID)
- Handle missing values (Median Imputation)
- Feature Scaling (StandardScaler)
- Train-Test Split (Stratified)
- Model Training (Linear SVM)
- Hyperparameter Tuning (GridSearchCV)
- Model Evaluation
- Visualization of Results

Workflow Diagram:

Dataset
↓
Data Cleaning
↓
Imputation
↓
Feature Scaling
↓
Train/Test Split
↓
Model Training (SVM)
↓
Hyperparameter Tuning
↓
Evaluation (Accuracy, ROC, etc.)

↓
Visualization

Performance Analysis

Evaluation Metrics Used:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC-AUC Curve

Interpretation:

- **Accuracy (~80–83%)** → Overall prediction correctness
- **Precision** → How many predicted defaulters were correct
- **Recall** → How many actual defaulters were identified
- **F1-score** → Balance between precision and recall
- **ROC-AUC** → Model's ability to separate classes

Since this dataset is imbalanced:

F1-score and Recall are more important than Accuracy

In financial risk modeling:

- Higher Recall reduces missed defaulters
- Balanced Precision avoids false alarms

Hyperparameter Tuning

Tuned Parameters:

Parameter	Description
C	Regularization strength
class_weight	Handles imbalance
max_iter	Convergence control

GridSearch Used:

C = [0.1, 1, 10]

- Cross-validation: cv = 3
- Scoring metric: F1 Score

Impact of Tuning:

- Improved F1-score
- Reduced overfitting
- Balanced bias-variance tradeoff
- Better minority class detection

Final Conclusion

- SVM is effective for structured financial datasets.
- LinearSVC performs well on large datasets.
- Hyperparameter tuning significantly improves generalization.
- Feature scaling and class balancing are essential for optimal performance.
- The model successfully predicts credit default with strong classification performance.

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Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.98      0.98        42
           1       0.99      0.99      0.99        72

    accuracy                           0.98       114
   macro avg       0.98      0.98      0.98       114
weighted avg       0.98      0.98      0.98       114
```

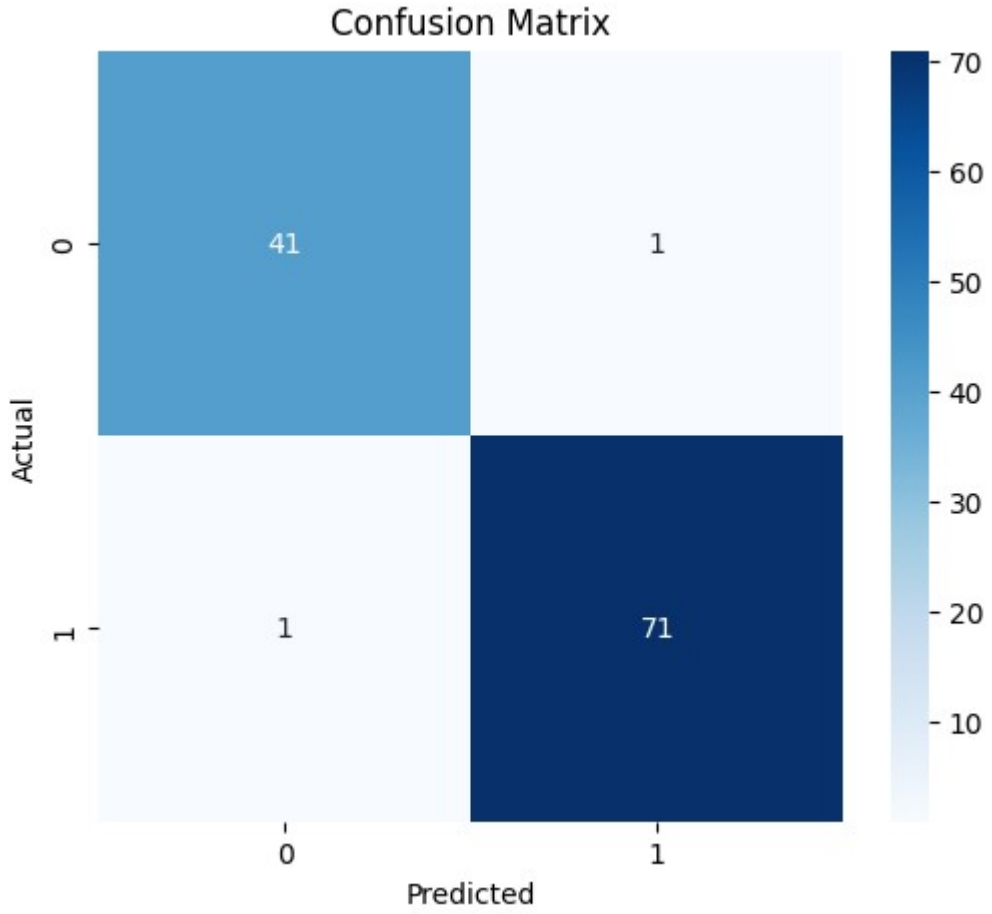

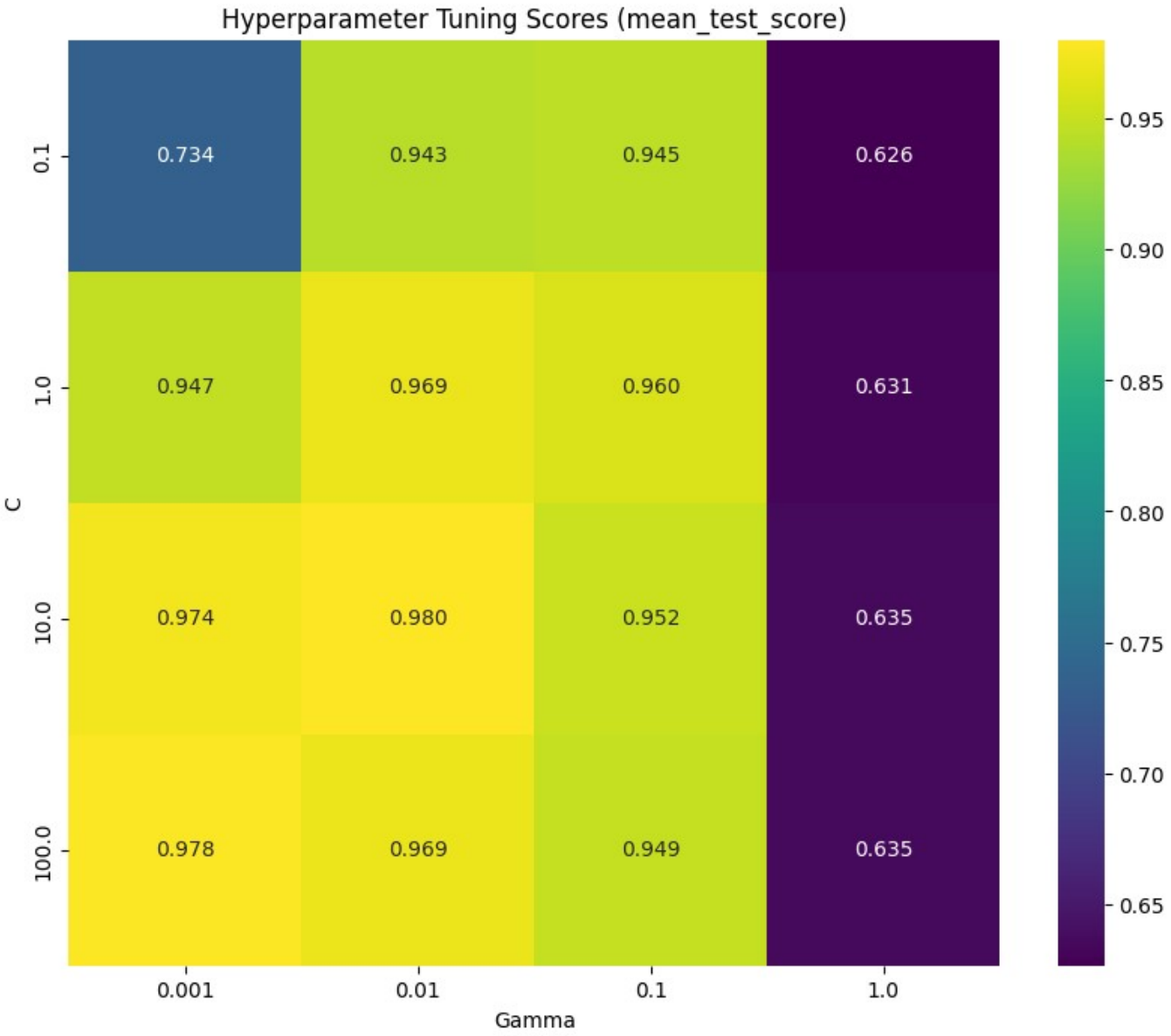
Hyperparameter Tuning Scores (mean_test_score)
C
0.1
1.0
10.0
100.0
0.734
0.943
0.945
0.626
0.947
0.969
0.960
0.631
0.974
0.980
0.952
0.635
0.978
0.969
0.949
0.635
0.001
0.01
0.1
1.0
Gamma
0.95
0.90
0.85
0.80
0.75
0.70
0.65